
Queries on your paper and its implications for England's PISA scores

9 messages

Tom Piccolo <tompiccolo50@gmail.com> 3 February 2024 at 21:22
To: Benton.t@cambridgeassessment.org.uk, Carroll.m@cambridgeassessment.org.uk

Hello

I hope you are well.

I have read your paper "How do GCSE grades link to PISA scores?" and wanted to ask a few questions concerning its content.

The first question is regarding your prediction for England should everyone study single science. You predicted that the UK's performance would rise to be close to that of South Korea.

What score would this be? What PISA level of proficiency would students in England be performing at?

And also, does this prediction also assume that all conditions for all students are equally favourable?

If not, then what would the outcome be in terms of scores? It would obviously be even higher, but how much higher? Where would it place us in the PISA rankings? What would our score be? (Of course don't answer this question if it makes that assumption already)

My second question is concerning how the conclusions from your study may have been verified as correct by data I myself requested.

I recently requested the PISA 2018 scores in Maths for those pupils who went onto achieve an Attainment 8 of 86, or 5 9's.

These pupils fall in the 95th percentile for GCSE results, nationally.

According to research by John Jerrim, a UCL statistician and researcher, if a representative sample for England was chosen for PISA 2018, then the minimum score that this group should be attaining in PISA Maths is 641, as the 95th percentile of PISA scores in this representative sample would be 641.

This should approximate to the 95th percentile of GCSE results in a representative sample.

Therefore, I was quite shocked to find that, to the contrary, the score for 86 was actually 612. Nowhere near 641.

For a representative sample in PISA 2018, the 90th percentile would be 610 (~ 612). This does not fit, at all, with national statistics and suggests the top 10% of pupils in England have 5 9's or more.

Actually, only the top 1-2% of pupils in England have 5 9's or more!

So, again taking this representative sample, they should be in the top 1-2% of England's PISA scores! Right?

You can see the point I am trying to make. 612 as a PISA score is too low, not even fitting with our national statistics. The score seriously underestimates their ability.

I believe looking at PISA scores for Single Science students only will solve this problem.

Here is why.

Double science results in **a underestimation of student ability** in PISA, as students in England (even those at high levels of achievement) are unable to fully show what they can do on PISA as a result of the reduced curriculum.

Looking at PISA scores for Single Science students **only will therefore** give a much more accurate picture of what PISA level of proficiency- and what score- 5 9's represents (referring to the new 9-1 GCSE grades).

This is because:

They have the full Science curriculum, in contrast with Double Science students who have content simplified and cut out in several places. They are therefore not disadvantaged i

And it isn't about the Single Science pupils being more able, either.

If we control for GCSE attainment, we can first look at pupils with an Attainment 8 of 86 in the whole cohort.

Then we look at pupils with the same Attainment 8 (so no ability difference) but only **those who study Single Science**- and suddenly their score rises dramatically- because they can access more of PISA Maths- then that proves that their lower score was because their curriculum disadvantaged them and NOT because they were "more able" as students who are on the same level as them regarding GCSE grades did comparatively worse on PISA.

Therefore looking at PISA scores for different Attainment 8's with respect to Single Science students only is much more accurate.

The actual scores for pupils at each level of attainment- particularly at the higher end of the distribution- are much higher than what previous records show.

I believe looking at PISA scores with relation to A Level results will add further proof to this.

Students with grades of AAA or above have typically grades 8-9 in their GCSE's, including Single Science.

The interesting thing is this.

Pupils with similar- controlled levels of prior ability- but who take Double as compared to Single Science, face serious setbacks and disadvantages when it comes to A Level, with less than half of those with 8-8 or above in Double Science gaining A-A* at A Level. This is in contrast with the vast majority of those taking Single Science with 8's or above gaining A-A* at A Level.

What does this mean? That yet again, Double Science has disadvantaged pupils who otherwise could have gotten higher PISA scores (if they did Single Science) and higher A Level results.

Had their curriculum been different.

This is reflected in your prediction for England, in that PISA scores and therefore also GCSE and A Level attainment would rise, if nationally, Single Science became the norm everywhere.

Therefore, those students who do get grades A*AA and above at A Level should have a mean PISA Maths score of 641 or above, as they were in the 95th percentile of GCSE results (and hence PISA scores)

This should be particularly noticeable for those who did A Levels in STEM subjects, and other academic A Levels.

By means of common sense, their attainment would place them in the 95th percentile of GCSE results, and therefore due to the concept of GCSE to A Level progression their PISA scores should be within that range:

95th percentile of PISA scores for England ~ 95th percentile of GCSE results

Many thanks for reading this email and I hope to receive a response in due course.

Kind regards

Tom Piccolo

Tom Benton <tom.benton@cambridge.org> 6 February 2024 at 11:29
To: Tom Piccolo <tompiccolo50@gmail.com>

Hello Tom

Let me respond to your queries. Overall, I think you need to be a bit more careful in the way you interpret both our research and PISA scores in general.

In short:

- Our report was not really making a prediction. Just adding some colour to a discussion of differences between students entering different subjects.
- No, people in the top 1 or 2% in GCSE will not be in the top 1 or 2% for PISA scores due to the fundamentally different things they measure and the (very) different way the two metrics are generated. This is part of the well-known phenomena of regression towards the mean.
- PISA scores are a intended to measure performance for all students in a country not just those following a particular educational programme. So, no, I don't think it would be appropriate to restrict England's sample to those doing separate sciences only.

Some more notes below.

Hope they are clear.

Tom

A "prediction" for if everyone did separate sciences

Let's be clear that the "prediction" in our report was subject to some very explicit caveats about not confusing correlation and causation. So it isn't really a prediction at all. It's more included as an interesting way of thinking about the sizes of differences in PISA performance between people studying different subjects. As we state in our report, the biggest issue with using the calculations (explained below) as a prediction is that, in reality, we have no idea what would happen if students who really don't like sciences were forced to take 3 GCSEs in them. Given all this, I don't think answering questions about the assumptions of the "prediction" is sensible.

That said, in terms of how it works, Table 4 of our report (https://www.cambridgeassessment.org.uk/Images/517876-the-link-between-subject-choices-and-achievement-at-gcse-and-performance-in-pisa-2015.pdf) indicates that performance in PISA is about 25 points, 20 points and 25 points higher for those that studied separate science than for those that do not (all other variables being equal). In 2015, roughly two-thirds of students in England were studying combined science GCSE rather than separate sciences so you could make a very rough and naïve prediction about the impact of moving students to separate sciences by multiplying the above coefficients by 0.67 and adding the values to existing average performance in England (e.g., 25*0.67=17 extra points in reading).

You can look up the scores for all countries in PISA 2015 in this report (https://www.oecd-ilibrary.org/education/pisa-2015-results-volume-i_9789264266490-en - reports for more recent reports are also available online). For example, Table I.4.3 shows the mean scores overall for the UK and for Korea in reading. If you want specific results for England then you can look at tables like Table B2.I.6 (though the results tend to be very similar to those for the UK as a whole). England were at 500 for reading so add 17 (see above) and we get to 517 which is the same score as Korea. You can do similar calculations for Science and Maths and "predict" that England would come out a bit ahead of Korea in science and a bit behind in Maths.

PISA scores and attainment 8

In short, the issue you are having is just regression to the mean, exacerbated by the fact that PISA scores can't be treated like normal test scores and won't correlate that highly with things like attainment 8. We went to a lot of trouble in our report (see pages 6 to 9) to address this issue before looking at the relationship between GCSE and PISA scores. I'll try and explain a little further.

PISA scores are recorded in the form of plausible values. Effectively these represent reasonable guesses of where each student's ability in each subject might be given what we know about them. Note that, when these are generated by OECD neither attainment 8 nor any part of GCSE performance is used in their creation. Plausible values are not designed to be accurate for individual students but rather so that, when viewed as a whole across large sample of students, they will give an accurate picture of the distribution of ability at a national level. This means, for example, that you might have PISA Maths scores (as plausible values) for students that did not even take any maths questions within the PISA tests.

Plausible values are used because the PISA tests themselves are quite short and not suitable for providing scores for individual students. Therefore, some very technical processes are required to extrapolate the overall national ability distributions in a way that accounts for the uncertainty in the tests themselves.

In the end, this means that the correlation between PISA scores for individuals and external variables like attainment 8 will be fairly low. In our report, we found the correlation of the original PISA maths scores with mean GCSE grade (the closest thing we had to attainment 8) was 0.68 (table A3).

This brings us to regression to the mean. If the correlation between two variables is 0.68, then, on average, people who are one standard deviation above the mean on one metric (say Attainment 8) will only be 0.68 standard deviations above the mean on the other (say PISA Maths scores). So:

- Let's say the 95th percentile of PISA Maths performance is 645.
- This is 1.6 standard deviations above the mean of 493 in 2015 when the standard deviation was 95 (i.e., 645=493+1.6*95).
- Assuming a correlation of 0.68 between attainment 8 and Maths GCSE scores, the average scores for people at the 95th percentile on attainment 8 would only be 596(=493+1.6*95*0.68).

So, whilst getting exactly to the bottom of your results would be quite complicated (going via the exact mechanisms of how PISA plausible values are generated and how strongly they correlate with attainment 8 in 2018), your results don't surprise me.

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Tom Piccolo <tompiccolo50@gmail.com> 6 February 2024 at 11:50
To: Tom Benton <tom.benton@cambridge.org>

Hello

I wanted to ask some questions:

You cannot use PISA to say "People with 5-6 9's are inferior to high performers in Korea" as their PISA score is relatively low; because of the issues you raised with PISA unable to be treated like a normal assessment?

What if when I do get the results for Single Science linked to Attainment 8, the results on PISA Maths are significantly higher?

(For people with the same Attainment 8, they have a higher PISA Maths score due to Single Science replacing Double)

Let me clarify that I have not obtained the PISA scores by Attainment 8 for Single Science.

Also, doesn't Double Science mean that student ability in England is underestimated? Didn't you state this yourself in the report?

Also you said the following:

So, no, I don't think it would be appropriate to restrict England's sample to those doing separate sciences only

But people doing Double Science have been underestimated by PISA: they form the bulk of the student cohort. So 612 is simply not reflective of a student who gets 5 grade 9's

Also, both PISA and GCSE results correlate strongly with IQ/general intelligence at Rs values of 0.83 and 0.81 respectively, so they do measure similar traits

Many thanks

Tom Piccolo
[Quoted text hidden]

Tom Piccolo <tompiccolo50@gmail.com> 6 February 2024 at 12:17
To: Tom Benton <tom.benton@cambridge.org>

Hello,

Also, what if when I do get the results for Single Science students we find out that the below statement is not true?

the average scores for people at the 95th percentile on attainment 8 would only be 596(=493+1.6*95*0.68)

What if it is not 596 but much higher? What if people who have the same Attainment 8 **but do Single Science** score much better? This means that the correlation between GCSE and PISA would rise AND that the scores we have are underestimates.

I asked for the mean PISA Maths score for 2018 for those who did Single Science only.

If it was higher, I think it would mean that PISA would better capture their ability than those who did Double Science.

Therefore it would mean that doing Single Science enables things to be "captured" in PISA that are not recorded for people doing Double Science (the majority)

We know that Double Science offers a reduced curriculum so you can't use PISA scores for (the majority of students) who are Double Science to estimate ability **as you get an underestimation of their ability**

Many thanks

Tom Piccolo

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Tom Benton <tom.benton@cambridge.org> 6 February 2024 at 12:23
To: Tom Piccolo <tompiccolo50@gmail.com>

Replies below.

I now need to do some other work.

Tom

From: Tom Piccolo <tompiccolo50@gmail.com>
Sent: Tuesday, February 6, 2024 11:51 AM
To: Tom Benton <tom.benton@cambridge.org>
Subject: Re: Queries on your paper and its implications for England's PISA scores

Hello

I wanted to ask some questions:

You cannot use PISA to say "People with 5-6 9's are inferior to high performers in Korea" as their PISA score is relatively low; because of the issues you raised with PISA unable to be treated like a normal assessment?

You can say this if you like but you'd need to be really explicit about what you mean by this and how you've calculated it. I wouldn't calculate it by looking at the average PISA scores of people with 5-6 9s. That won't work due to the regression towards the mean issue and the complicated way PISA scores work.

But you could with effort make an argument like (and I'm making the following up) "people with 5-6 9s are in the top 1% of the country, the top 1% of performers in PISA display performance have scores of XXX which is superior to the average performance in Korea. Thus, in some sense, students achieving 5-6 9s demonstrate a level of superiority over other students in England far in excess of the difference in average performances between England and Korea in the PISA tests". Not saying you should make an argument like this but you could.

What if when I do get the results for Single Science linked to Attainment 8, the results on PISA Maths are significantly higher? That would be as expected and would support the idea that those who do separate sciences do better on PISA tests (as we say in our research). I think the thing to avoid is assuming that this means that forcing other students to study separate science would necessarily boost *their* scores (correlation and causation). Also, just need to be careful not to overinterpret the numbers you get as the way they are generated is quite complicated. A simple conclusion saying those that study separate sciences tend to have higher PISA scores is fine.

I response to your next email. The calculations I've shown are just an example of what regression to the mean *might* do. I'm not expecting you to see exactly the same results in your own data. I just want you to be aware that people in the top 1-2% on one metric won't be in the top 1-2% on another. This is completely normal and not something to overinterpret.

(For people with the same Attainment 8, they have a higher PISA Maths score due to Single Science replacing Double)

Let me clarify that I have not obtained the PISA scores by Attainment 8 for Single Science.

Also, doesn't Double Science mean that student ability in England is underestimated? Didn't you state this yourself in the report? No I don't think we've said this anywhere.

Also you said the following:

So, no, I don't think it would be appropriate to restrict England's sample to those doing separate sciences only

But people doing Double Science have been underestimated by PISA: they form the bulk of the student cohort. So 612 is simply not reflective of a student who gets 5 grade 9's

Not sure I follow but people doing double science haven't been underestimated. They just don't do as well as those studying separate sciences. That's not the same.

Also, both PISA and GCSE results correlate strongly with IQ/general intelligence at Rs values of 0.83 and 0.81 respectively, so they do measure similar traits

Yes similar but not the same. And it will depend what you mean by "PISA results". As I've said, the numbers you get in the data are plausible values. The exact correlations between these and GCSE results are given in our report (so just 0.68 for average GCSE performance and PISA Maths plausible values).

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Tom Piccolo <tompiccolo50@gmail.com> 6 February 2024 at 12:38
To: Tom Benton <tom.benton@cambridge.org>

Hello

Just one more thing:

People who do Double Science have content significantly reduced and cut out. Surely this would mean that their PISA score is an underestimate of their ability? So 612 is too low for 86, in terms of representing what an Attainment 8 of 86 actually is.

—> People with the same Attainment 8- the same controlled academic ability- but who do Single Science, do much better on PISA? So "86" attainment 8 or 5-6 9's is actually a much higher value on PISA than previously reported

Many thanks

Tom Piccolo
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Tom Piccolo <tompiccolo50@gmail.com> 6 February 2024 at 12:49
To: Tom Benton <tom.benton@cambridge.org>

Hello

Please see below for the point I was trying to make:

If we control for GCSE attainment, we can first look at pupils with an Attainment 8 of 86 in the whole cohort.

Then we look at pupils with the same Attainment 8 (so no ability difference) but only **those who study Single Science**- and suddenly their score rises dramatically- because they can access more of PISA Maths- then that proves that their lower score was because their curriculum disadvantaged them and NOT because they were "more able" as students who are on the same very high level as them regarding GCSE grades did comparatively worse on PISA.

So my question is: if people with an Attainment 8 of 86 who do Single Science can get very high PISA Maths scores, **then what is stopping the other people in that group who get the same Attainment 8 of 86 from doing the same thing?**

If they had a better curriculum. The ability of these students with the same Attainment 8 is the same remember but their curriculum differs- one is much more cut out and reduced

Many thanks

Tom Piccolo
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Tom Piccolo <tompiccolo50@gmail.com> 6 February 2024 at 12:56
To: Tom Benton <tom.benton@cambridge.org>

Hello

I think I didn't get across my point well enough; please see the above two emails I sent you which better explain what I was trying to say.

Many thanks

Tom Piccolo
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Tom Piccolo <tompiccolo50@gmail.com> 6 February 2024 at 15:14
To: Tom Benton <tom.benton@cambridge.org>

HI Tom

I hope you are well

My question I believe has been confused quite significantly in my wording of it in my previous emails. I will try and be as concise as possible in this email.

When I mentioned the underestimation of Double Science, what I meant was this:

We know pupils who do Single Science and do PISA (e.g PISA Maths) score much higher than those who do not. I mentioned this means they have been underestimated.

I believe there may have been a confusion as to what I meant by this. You said that switching these students to Single Science would not necessarily boost their scores.

This may be true for the lower levels of attainment at GCSE, but what about those who have, say, 5-6 9's at GCSE?

Essentially the following situation arises:

The group who gets 5-6 9's or more at GCSE:

Some of them do Single Science
The rest do Double Science*

* the curriculum of Double Science is significantly simplified and reduced in many places

But, the academic ability of the students in this group who get 5-6 9's or higher is more or less the same; however a subsection of this group has substantially higher PISA scores (those who do Single Science)

There is a difference in the curriculum between the two.

So therefore, curriculum differences mediate the lower PISA scores for this already very high attaining GCSE group. So therefore that is why I am proposing:

If pupils who have an Attainment 8 of 86 (5 9's) who do Single Science have such a high PISA score, then why can't the other people in that group (who have the same attainment 8 and the same GCSE attainment) do the same if they switched?

Why couldn't the top performers in GCSE who do Double Science do as well (in PISA) as the top performers who do Single Science?

Their ability is the same.

I hope my point has been clarified by this email

Hope to receive a response in due course,

Tom Piccolo
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